Formal Specification of the Normal Distribution

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

keith.lohse@health.utah.edu

@keith_lohse

HEALTH
UNIVERSITY OF UTAH

“Know thy Distributions”

- It’s easy to get lost in the weeds of mathematical formalisms.
- But it’s important to dwell here for a while so that we appreciate the power that understanding these distributions allows.
- Specifically, knowing a distribution allows us to make quantitative statements about the probability of an event.
  - I.e., “Assuming a normal distribution, what is the probability of observing a values >/= 1?”

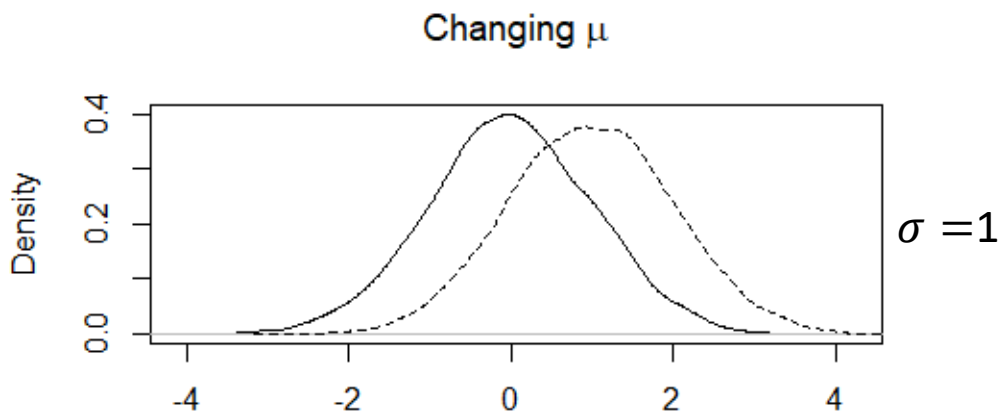
The Normal Distribution

- $f(x) = \frac{1}{\sigma(2\pi)} e^{-(x-\mu)^2/2\sigma^2}$

- Where μ is the mean and σ is the standard deviation. When $\mu = 0$ and $\sigma = 1$, this is referred to as the standard normal distribution.

The Normal Distribution

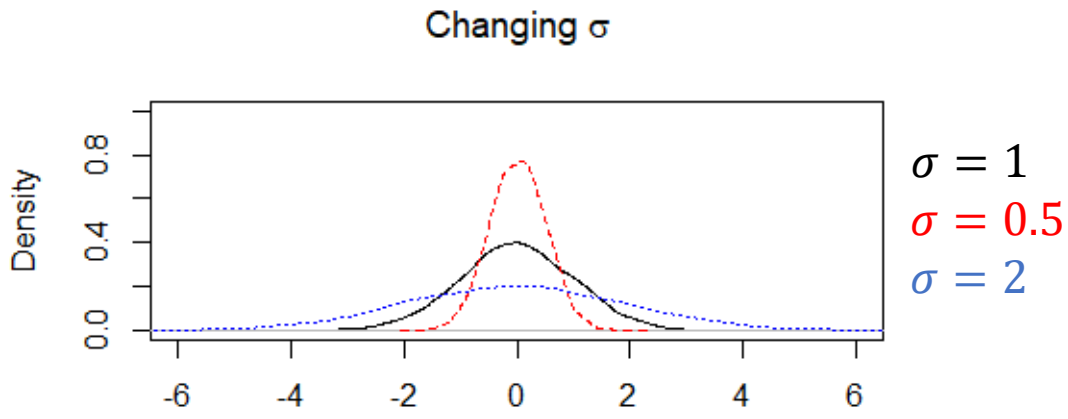
- $f(x) = \frac{1}{\sigma(2\pi)} e^{-(x-\mu)^2/2\sigma^2}$



$\sigma = 1$

$\mu = 0$

$\mu = 1$



$\sigma = 1$

$\sigma = 0.5$

$\sigma = 2$

$\mu = 0$

The Normal Distribution

- For any of these normal distributions, we can find the probability that a number lies in a specific range (e.g, between -1 and +1, or greater than +2).

- Technically, as our values are continuous, the probability of being precisely one is zero, p(1) = 0. This is because, for a continuous variable, 1 is different from 0.999 and 1.000.
  - With infinite precision, the probability of something being ***exactly*** anything is going to be 0.
  - However, we could look in a very narrow range (1.00 to 1.01).

Standardizing a variable

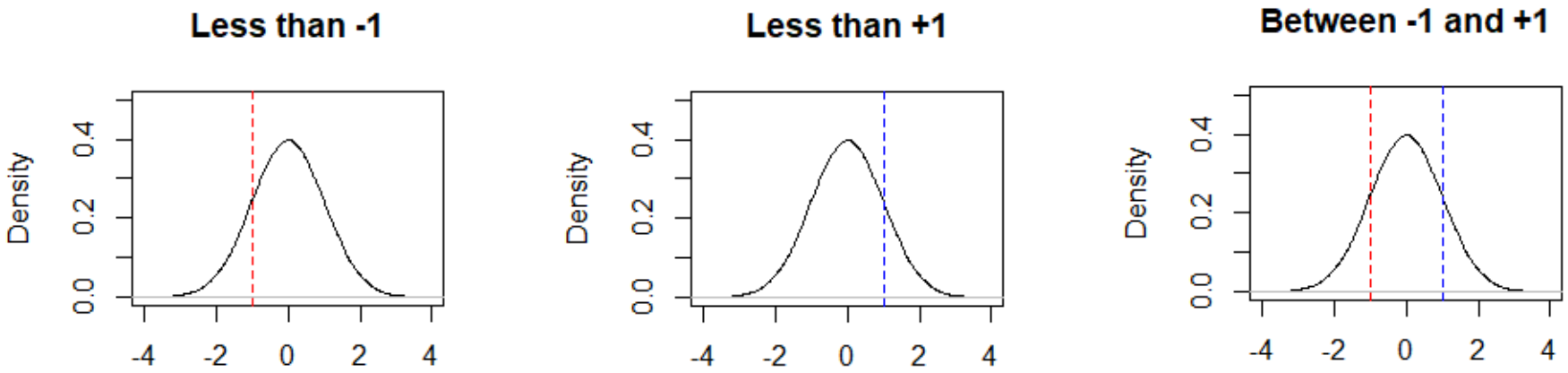
- All of these normal distributions are different, thus the probability of being within -1 to +1 changes depending on μ and σ.

- However, we can standardize any value of Y by transforming it into a new value called a ***z-value*** or z-score.
  - $z = \frac{Y-\mu}{\sigma}$

- Thus, for $Y = 1$, when $\mu = 0, \sigma = 1$, the z-value is 1.
- However, when $\mu = 0, \sigma = 0.5$, the z-value is 2!

Standardizing a variable

- Once standardized, we can find the probability associated with specific z-values!



15.8% ***84.1%*** ***68.3%***

(We will use built-in functions in R to calculate the probabilities for continuous variables, but we will calculations for some discrete variables by hand!)

Know thy distributions

- We will discuss how to calculate these probabilities in more detail for both normal and non-normal distributions.

- Regardless of the specific distribution, the lesson is the same: if we know the underlying distribution or we are confident the distribution is a reasonable assumption, we can make very explicit quantitative statements about the probability of observing an event.